

Vivek M G

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OBJECTIVE

Applying for admission to the MS in Computer Engineering program at the National University of Singapore for Fall 2026.

EDUCATION

B Tech in Computer Science and Engineering , SRMIST, Kattankulathur	CGPA- 9.26/10	May 2026*
12th Grade , Kola Perumal Chetty Vaishnav Senior Secondary School, Chennai	70.4%	Jul 2022
10th Grade , National Public School, Chennai	78.8%	Jun 2020

RESEARCH INTERESTS: Embedded AI, Internet of Things (IoT), Artificial Intelligence (AI)

COURSEWORK TAKEN: Computer Organization and Architecture, Artificial Intelligence, Internet of Things (IoT), Embedded System Design Using Arduino, Machine Learning (ML)

INTERNSHIPS

Research and Development Intern, Samsung R&D Institute India

Mar 2024 – Dec 2024

Detecting Anomaly in Smart Home Events using GenAI

- Designed a smart lock anomaly detection system using LLAMA2-7b Chat for event classification and isolation forest for detecting anomalies, enhancing home security by proactively identifying unusual access patterns.
- Created a synthetic dataset using Python & SmartThings API for simulating realistic lock/unlock activity, ensuring robust model training despite limited real-world data.
- Pre-processed data to reduce bias and train ML models to learn user patterns and visualized event trends using Matplotlib library in Python, improving model fairness and providing actionable insights into user behavior.
- Evaluated the preprocessed dataset with five unsupervised learning algorithms; isolation forest outperformed other unsupervised learning algorithms in anomaly detection; used LLAMA2-7b to generate real-time natural language explanations, delivering explainable AI alerts to increase user trust and system usability.
- Achieved 90% accurate real-time anomaly identification with <100ms latency, enabling faster responses to potential security breaches.

UNDERGRADUATE PROJECT

Integrating IoT Health Monitoring Devices in Automotive Vehicles

Dec 2025 – April 2026

- Extended the minor project to an automotive setting by integrating Arduino Nano ESP32 with an A7670E Cat-1 GSM module, bridging BLE 33 sensor data via UART for real-time cloud transmission with live GPS tracking, enabling continuous health monitoring across cellular networks.
- Incorporated IMU-based inactivity detection to trigger a vibration coin motor after 30 minutes of no movement, alerting drivers to prolonged static posture to encourage circulation and reduce sitting-related discomfort.
- Built VitalFlow, a cross-platform mobile app using React Native and Supabase, providing a live health dashboard with time-windowed analytics, severity-based alert generation, and push notifications for caregivers.
- Implemented an SOS emergency feature with GPS reverse geocoding to auto-detect the national emergency number and dispatch the driver's location and health snapshot to registered contacts via GSM SMS.
- Achieved an average cloud API response time of ~390ms, enabling near real-time physiological data delivery from vehicle to cloud over a 4G cellular network.

Embedded Machine Learning for Early Heart Attack Prediction Symptoms

Jan 2025 – May 2025

- Designed a real-time heart attack prediction system using Arduino Nano BLE 33 Sense Rev 2 and Embedded AI, making predictive health analytics possible on small, resource-constrained devices.
- Integrated MAX30102 (Heart rate and SpO₂), TMP117 (body temperature) and GSR (Skin Resistance) sensors to collect physiological signals, capturing multi-modal data for more reliable symptom detection.

- Created a synthetic dataset simulating heart attack symptoms using Python, NumPy, Pandas and statistical sampling, performed preprocessing, labeling and visualization using Matplotlib, addressing the scarcity of annotated medical datasets.
- Trained a binary classification model using a TensorFlow Lite neural network to address the lack of real-time medical sensor data; Achieved 96% accuracy with ongoing improvements to reduce bias.

COMPETITION PROJECTS

Enduraverse 25, Endurance Technologies

Low-Cost Telematics Dongle

18 Feb 2025 – 20 Feb 2025

- Designed and built a low-cost telematics dongle using ESP32 and an IMU sensor to monitor real-time acceleration, gyroscope, direction, speed, and jerk, enabling accurate motion tracking for automotive and industrial applications.
- Handled the complete hardware implementation and developed the mobile application for Bluetooth-based data transmission; integrated ThingSpeak for cloud-based data logging and graphical visualization of motion data, enabling seamless remote monitoring and faster analysis of vehicle performance.
- Implemented the product with a focus on low latency, achieving 85 ms real-time response compliant with company specifications, ensuring timely and reliable motion data for automotive and industrial applications.
- Designed the system to support future applications such as estimating part lifetime and mechanical wear by analyzing long-term motion patterns from sensor data, providing a foundation for predictive maintenance and reducing operational costs.

ULTRON 8.0 FUTURIX.CTECH, SRMIST

IoT based vision glasses for the blind

30 Jan 2025 – 31 Jan 2025

- Developed an IoT-based vision aid using Jetson Nano and MiDaS depth estimation to detect obstacles and provide real-time spatial feedback, enhancing mobility and safety for visually impaired users.
- Developed a Bluetooth-connected mobile app delivering audio and haptic alerts for intuitive environmental awareness, allowing users to navigate unfamiliar spaces independently.
- Achieved 87% obstacle detection accuracy in real-time testing during the hackathon prototype, demonstrating the system's practical effectiveness in real-world scenarios.

COURSE PROJECTS

Robotic Arm Using ESP32

Mar 2024 – Apr 2024

- Designed a robotic arm using ESP32-WROOM with Wi-Fi control, WebSocket communication, and PWM-based movement.

Road Damage Detection using Detectron 2

Jan 2024 – Apr 2024

- Designed and trained a computer vision model using Detectron 2 for detecting road damage such as potholes and cracks; designed a Flask backend for serving model predictions and a frontend using HTML and Tailwind CSS
- Achieved 85% accuracy on city road datasets through proper preprocessing and augmentation

Beyond Pages Trust Website

May 2024 – Jul 2024

- Designed a website using HTML and Tailwind CSS with Razorpay integration for donation processing

Blog-App

Jan 2024 – Apr 2024

- Built a real-time blogging platform using React.js, Prisma with MongoDB for structured data management, and Firebase for authentication and live updates.

CERTIFICATIONS

- GitHub Foundation, GitHub, Feb 2025
- Networking Basics, Cisco Networking Academy, Cisco, Sep 2024
- Operating System Basics, Cisco Networking Academy, Cisco, Mar 2024
- Introduction to IoT, Cisco Networking Academy, Cisco, Mar 2024
- Computer Architecture, IIT Madras, NPTEL, Oct 2023
- Programming in Java, IIT Kharagpur, NPTEL, Oct 2023

TECHNICAL SKILLS

- **Operating Systems:** Windows, Linux
- **Languages:** C++, Python, JavaScript, Arduino C

- **Databases:** MySQL, MongoDB, Supabase
- **Frameworks & Libraries:** React Native, React JS, Next JS, TensorFlow Lite
- **Software Tools:** VS Code, GitHub, Git, Figma
- **Others:** Embedded Systems Programming, IoT Development

AWARDS AND EXTRA-CURRICULAR ACTIVITIES

- Placed 3rd among 6 teams, Enduraverse'25 Hackathon, Endurance Technologies Ltd, Feb 2025; developed a low-cost telematics dongle
- Led a 4-member design team for the Creatives domain of Alexaverse, the flagship tech event organized by Alexa Developers SRM, held at SRMIST, Kattankulathur; personally designed the event poster and established a consistent visual identity, significantly enhancing branding and attendee engagement.
- Volunteer, Beyond Pages Trust: Developed and maintained the NGO's website, organized fundraising initiatives, taught Computer Science to 11th-grade state board students, and coordinated grocery and clothing donation drives to support rural communities, Jun 2024 – Jul 2025.
- Logistics Volunteer, GIT101: Handled troubleshooting for a hands-on session on Git and GitHub conducted at SRMIST, Kattankulathur, for 250+ attendees across all Engineering Departments, Nov 2023.
- Completed Trinity College London Electronic Keyboard examinations from Initial to Grade 6 (2011–2020), reflecting long-term dedication, discipline, and advanced musical proficiency.

LANGUAGES

- English, Tamil (Read/Write/Speak)